Conceptual physics (CP) frame:
Students and instructor are in this frame when they discuss physics scenarios or phenomenon, about properties of physics quantities related to the task at hand. They may also exploit the symmetry of a physical system by investigating related concepts.

Algorithmic physics (AP) frame:
In the algorithmic physics frame, students recall physics equations or apply physics knowledge to rearrange known equations using math. Students may also derive expressions for specific cases from a general physics equation or validate an expression via dimensional analysis.

Algorithmic math (AM) frame:
This frame refers to performing mathematical computation by following well-established protocols without questioning the validity of those protocols, e.g., solving an equation or computing an integral.

Conceptual math (CM) frame:
Students are in this frame when they exploit properties of mathematical constructs to quickly obtain a result without diving into algorithmic manipulation, e.g., noticing that all the odd terms in a sum are equal to zero.

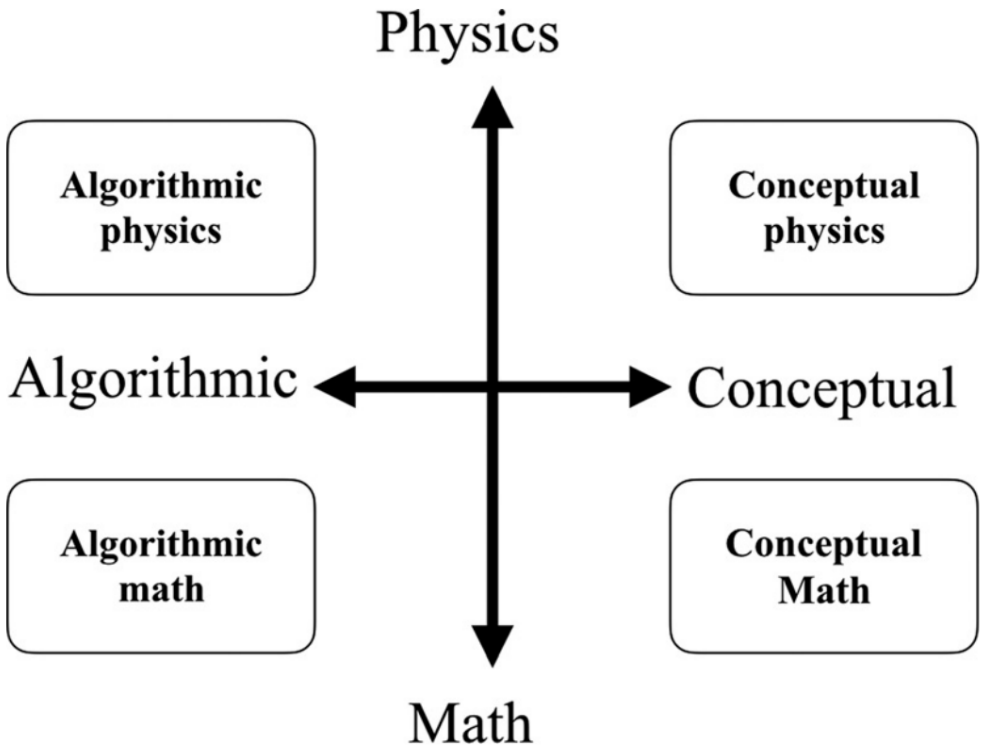

Figure and text from: *Am. J. Phys.* 87, 875–884 (2019)